

# The Koide Constant Decomposes

## Generation Count and Amplitude in the Charged Lepton Mass Relation

**Registry:** [HOWL-PHYS-8-2026]

**Series Path:** [HOWL-PHYS-1-2026] → [HOWL-PHYS-5-2026] → [HOWL-PHYS-7-2026] → [HOWL-PHYS-8-2026]

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## I. ABSTRACT

The Koide formula (1981) states that the three charged lepton masses satisfy  $(m_e + m_\mu + m_\tau) / (\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$ , confirmed to 0.0009% by current measurements. The constant 2/3 has remained unexplained for 45 years. This paper decomposes it.

For  $N$  equally-spaced objects on a circle in  $\sqrt{(\text{mass})}$  space with modulation amplitude  $a$ , the mass ratio  $(\Sigma m) / (\Sigma \sqrt{m})^2 = (1 + a^2/2)/N$ . This is a trigonometric identity valid for  $N \geq 3$ , derived from two standard results:  $N$  equally-spaced cosines sum to zero, and the sum of their squares equals  $N/2$ . The identity is proven in Section III. The  $N = 2$  case is degenerate — the double-angle cosines do not cancel — and the identity does not hold for  $N = 2$ .

The Koide constant 2/3 is this identity evaluated at  $N = 3$  and  $a^2 = 2$ : the ratio becomes  $(1 + 2/2)/3 = 2/3$ . The denominator 3 is the number of charged lepton generations. The numerator 2 is the amplitude squared. Neither alone produces 2/3. Both contribute.

The amplitude  $a = \sqrt{2}$  is equivalent to the Koide relation. It is not derived from a separate physical principle. The observation that  $a = \sqrt{2}$  allows the parametrization to produce zero mass (when  $\cos = -1/\sqrt{2}$ ) provides physical motivation for why  $a^2 = 2$  is natural, but this does not constitute an independent derivation. The amplitude carries content equivalent to the formula itself.

What is new in this paper: the general formula  $(1 + a^2/2)/N$  for arbitrary  $N \geq 3$  and arbitrary amplitude  $a$ , the decomposition of the Koide constant into generation count and amplitude, and the computation of  $m_\tau$  from  $m_e$  and  $m_\mu$  in controlled-precision

rational arithmetic. The predicted  $m_\tau = 1776.97$  MeV versus PDG  $1776.86 \pm 0.12$  MeV — a tension of  $0.91\sigma$ , consistent with the formula being exact.

What is assumed: the Koide formula itself — that the three charged lepton masses are equally spaced on a circle in  $\sqrt{\text{mass}}$  space. This is empirical. It is not derived here. Why  $\sqrt{\text{mass}}$  is the natural space, why the spacing is equal, and what determines the free parameters  $M$  and  $\theta_0$  remain open questions stated here, not hidden.

In the framework of this series ([[HOWL-PHYS-1-2026](#)]), mass is inertia — resistance of a coherent pattern to disruption, not substance. The Koide formula is a geometric constraint on three vortex soliton inertias at maximum symmetry ( $120^\circ$  spacing) on a circle. The free parameter count of the Standard Model reduces from 18 (after [[HOWL-PHYS-7-2026](#)]) to 17, conditional on the formula being exact.

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## II. THE KOIDE FORMULA

### 2.1 The Relation

Yoshio Koide proposed in 1981 that the three charged lepton masses satisfy:

$$(m_e + m_\mu + m_\tau) / (\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$$

Using current PDG values ( $m_e = 0.51099895$  MeV,  $m_\mu = 105.6583755$  MeV,  $m_\tau = 1776.86 \pm 0.12$  MeV), the left side evaluates to 0.666661, deviating from  $2/3$  by 0.00092%. This deviation is  $0.91\sigma$  given the measurement uncertainty on  $m_\tau$ . The relation is consistent with being exact.

### 2.2 History

The formula has held for 45 years through multiple improvements in the measured value of  $m_\tau$ . At every stage, the Koide ratio has remained within measurement uncertainty of  $2/3$ . No measured improvement has moved the ratio away from  $2/3$ . It is the most precise empirical mass relation in particle physics.

### 2.3 The Two Open Questions

Two questions have been open since 1981.

First: why does the formula hold? What principle constrains the three charged lepton masses to satisfy this relation? This question remains open. This paper does not answer it.

Second: given that the formula holds, why is the constant  $2/3$  rather than some other value? This paper answers this question by decomposing  $2/3$  into two contributions: the generation count (3) and the amplitude ( $a^2 = 2$ ).

### III. THE GENERAL IDENTITY

#### 3.1 The General Parametrization

Three Charged Leptons on a Circle in  $\sqrt{m}$  Space

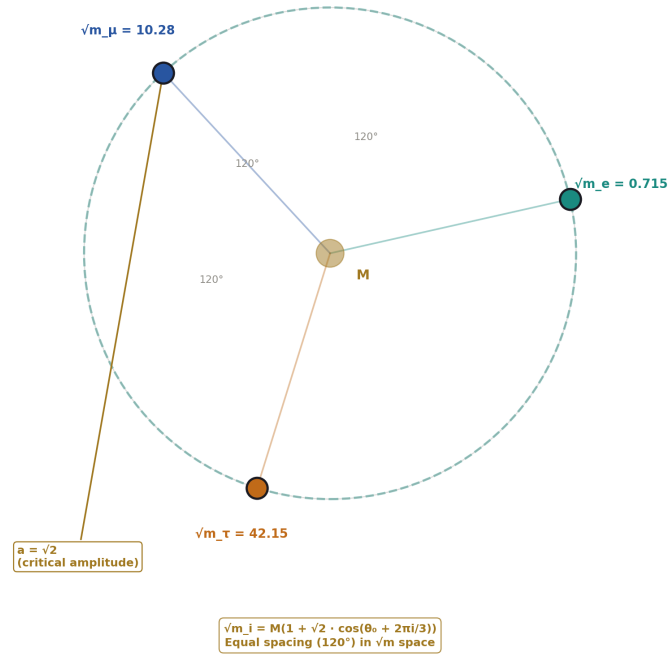


Figure 1: Fig. 1: Three charged leptons at  $120^\circ$  spacing on a circle in  $\sqrt{m}$  space — the Koide geometry with critical amplitude  $a=\sqrt{2}$  visible as the modulation radius.

Consider  $N$  objects with masses  $m_i$  parametrized as:

$$\sqrt{m}_i = M(1 + a \cdot \cos(\theta_0 + 2\pi i/N)) \text{ for } i = 0, 1, \dots, N-1$$

where  $M > 0$  is a scale parameter with dimensions of  $\sqrt{\text{mass}}$ ,  $a > 0$  is a dimensionless modulation amplitude, and  $\theta_0$  is an orientation angle. The  $N$  objects are equally spaced at angular intervals of  $2\pi/N$  on a circle in  $\sqrt{\text{mass}}$  space.

For  $N = 3$  and  $a = \sqrt{2}$ , this is the standard Koide parametrization.

### 3.2 The Sum of Square Roots

$$\sum \sqrt{m_i} = M \cdot \sum (1 + a \cdot \cos(\theta_0 + 2 \pi i/N)) = M \cdot (N + a \cdot \sum \cos(\theta_0 + 2 \pi i/N))$$

The sum of  $N$  equally-spaced cosines on a circle is zero for  $N \geq 3$ :

$$\sum_{k=0}^{N-1} \cos(\theta + 2 \pi k/N) = 0 \text{ for all } N \geq 3, \text{ all } \theta$$

This is a standard trigonometric identity. It follows from the fact that  $N$  equally-spaced unit vectors on a circle sum to the zero vector, and the cosine sum is the projection onto any axis.

**The  $N = 2$  exclusion.** For  $N = 2$ :  $\sum \cos(\theta + \pi k) = \cos(\theta) + \cos(\theta + \pi) = \cos(\theta) - \cos(\theta) = 0$ . The cosine identity does hold for  $N = 2$ . However, the  $\cos^2$  identity used below does not hold for  $N = 2$ , as shown in §3.4. The general formula is valid for  $N \geq 3$ .

Therefore  $\sum \sqrt{m_i} = NM$ , and  $(\sum \sqrt{m_i})^2 = N^2 M^2$ .

### 3.3 The Sum of Masses

$$\sum m_i = M^2 \cdot \sum (1 + a \cdot \cos(\varphi_i))^2$$

where  $\varphi_i = \theta_0 + 2 \pi i/N$ . Expanding:

$$= M^2 \cdot \sum (1 + 2a \cdot \cos(\varphi_i) + a^2 \cdot \cos^2(\varphi_i))$$

$$= M^2 \cdot (N + 2a \cdot \sum \cos(\varphi_i) + a^2 \cdot \sum \cos^2(\varphi_i))$$

The linear cosine sum is zero (for  $N \geq 3$ ). The  $\cos^2$  sum requires evaluation.

### 3.4 The Sum of Cosines Squared

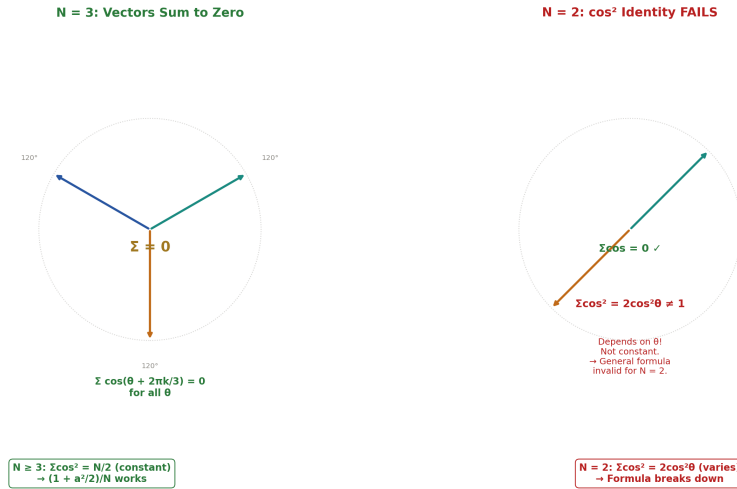


Figure 2: Fig. 4:  $N=3$  equally-spaced vectors sum to zero (left).  $N=2$   $\cos^2$  identity fails — depends on  $\theta$ , not constant (right). The general formula requires  $N \geq 3$ .

Using  $\cos^2(x) = (1 + \cos(2x))/2$ :

$$\Sigma \cos^2(\theta + 2\pi k/N) = N/2 + (1/2) \cdot \Sigma \cos(2\theta + 4\pi k/N)$$

The second sum is  $N$  cosines equally spaced at intervals of  $4\pi/N$ . For  $N \geq 3$ , these are still equally spaced around the circle (the spacing  $4\pi/N = 2 \cdot (2\pi/N)$  corresponds to going around the circle twice, but the  $N$  points remain equally spaced modulo  $2\pi$ ). Therefore the sum is zero for  $N \geq 3$ :

$$\Sigma \cos^2(\theta + 2\pi k/N) = N/2 \text{ for all } N \geq 3, \text{ all } \theta$$

**The  $N = 2$  failure.** For  $N = 2$ :  $\Sigma \cos(2\theta + 4\pi k/2) = \cos(2\theta) + \cos(2\theta + 4\pi) = 2\cos(2\theta)$ . This does not equal zero — it depends on  $\theta$ . The  $\cos^2$  identity fails for  $N = 2$ . The sum  $\Sigma \cos^2(\theta + \pi k) = \cos^2(\theta) + \cos^2(\theta + \pi) = \cos^2(\theta) + \cos^2(\theta) = 2\cos^2(\theta)$ , which depends on  $\theta$  and does not equal  $N/2 = 1$  except at  $\theta = \pi/4$ . The  $N = 2$  case is degenerate and excluded from the general formula.

### 3.5 The General Ratio

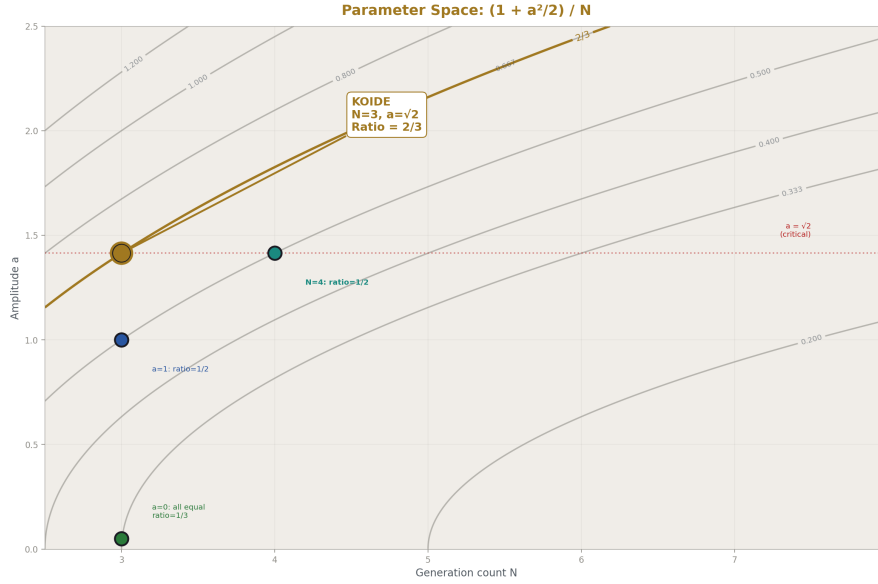


Figure 3: Fig. 5: Contour plot of  $(1+a^2/2)/N$  — the Koide point at  $N=3$ ,  $a=\sqrt{2}$  sits on the  $2/3$  contour. The full parameter space shows where other generation counts and amplitudes would land.

Substituting:

$$\Sigma m_i = M^2(N + 0 + a^2 \cdot N/2) = M^2 \cdot N \cdot (1 + a^2/2)$$

The ratio:

$$(\Sigma m) / (\Sigma \sqrt{m})^2 = M^2 \cdot N \cdot (1 + a^2/2) / (N^2 M^2) = (1 + a^2/2) / N$$

This is valid for all  $N \geq 3$ , all  $a > 0$ , all  $\theta_0$ , and all  $M > 0$ . The free parameters  $M$  and  $\theta_0$  cancel completely.

### 3.6 The Koide Evaluation

For the Koide formula:  $N = 3$ ,  $a = \sqrt{2}$ .

$$(1 + (\sqrt{2})^2/2) / 3 = (1 + 2/2) / 3 = (1 + 1) / 3 = 2/3$$

The constant decomposes:

- The denominator 3 is the generation count  $N$
- The numerator 2 is  $1 + a^2/2 = 1 + 1 = 2$ , determined by the amplitude  $a = \sqrt{2}$
- Together:  $2/3$

### 3.7 The Amplitude Question

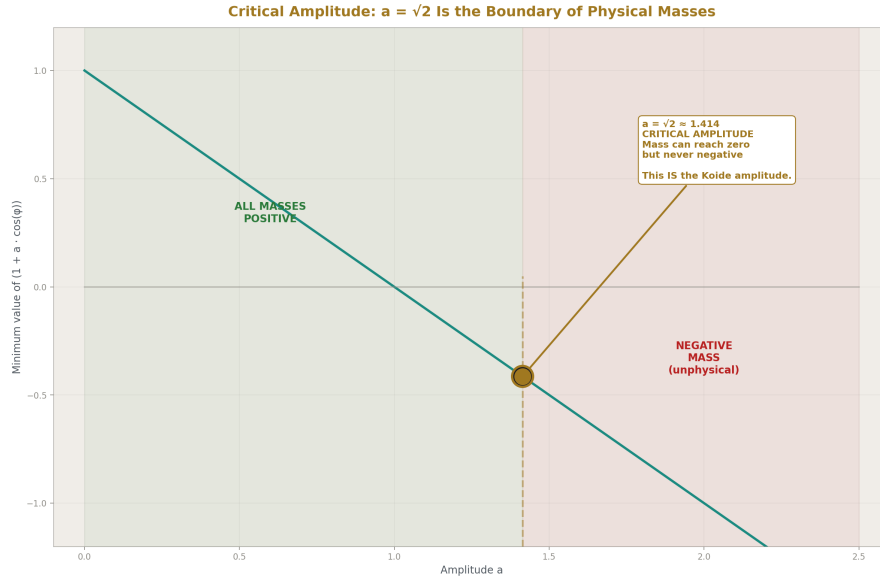


Figure 4: Fig. 3: The minimum mass factor vs amplitude — at  $a < \sqrt{2}$  all masses positive, at  $a = \sqrt{2}$  mass can reach zero, at  $a > \sqrt{2}$  masses go negative. The critical point IS the Koide amplitude.

The general ratio is  $(1 + a^2/2)/N$ . For different amplitudes at  $N = 3$ :

| Amplitude $a$ | $a^2$ | Ratio     | Simplified                    |
|---------------|-------|-----------|-------------------------------|
| 1             | 1     | $3/2 / 3$ | $1/2$                         |
| $\sqrt{2}$    | 2     | $2/3$     | $2/3 \leftarrow \text{Koide}$ |
| $\sqrt{3}$    | 3     | $5/2 / 3$ | $5/6$                         |
| 2             | 4     | $3/3$     | 1                             |

The value  $a = \sqrt{2}$  is not determined by the generation count. It is an independent input. The general formula shows that  $2/3$  requires both  $N = 3$  and  $a^2 = 2$ . Attributing  $2/3$  to the generation count alone would be an overstatement.

The amplitude  $a = \sqrt{2}$  is equivalent to the Koide relation. Setting the ratio equal to  $2/3$  and solving:  $(1 + a^2/2)/3 = 2/3$  gives  $a^2 = 2$ . Conversely,  $a = \sqrt{2}$  in the parametrization produces the Koide relation. The amplitude and the formula are two ways of stating the same constraint.

**Physical motivation for  $a = \sqrt{2}$ .** When  $a = \sqrt{2}$ , the factor  $(1 + \sqrt{2} \cdot \cos(\varphi))$  can equal

zero — this occurs when  $\cos(\varphi) = -1/\sqrt{2}$ , i.e.,  $\varphi = 3\pi/4$  or  $5\pi/4$ . This means the parametrization can produce  $m_i = 0$ . For  $a < \sqrt{2}$ , all masses are strictly positive. For  $a > \sqrt{2}$ , some masses become negative (unphysical). The value  $a = \sqrt{2}$  is the boundary between “all masses positive” and “some masses negative.” It is the maximum amplitude for which the parametrization produces physical (non-negative) masses.

This is a structural observation about the parametrization’s boundary. It explains why  $a = \sqrt{2}$  is special — it is the critical amplitude. It does not constitute a derivation independent of the Koide formula, because the critical-amplitude property and the Koide relation are equivalent statements about the same constraint.

### 3.8 The Generalization Table

For  $a^2 = 2$  (the Koide amplitude) at various  $N$ :

| N | Ratio $2/N$ | As fraction                     |
|---|-------------|---------------------------------|
| 3 | 2/3         | 0.667 ← Koide (charged leptons) |
| 4 | 2/4         | 0.500                           |
| 5 | 2/5         | 0.400                           |
| 6 | 2/6         | 0.333                           |
| 7 | 2/7         | 0.286                           |

If a Koide-type relation (with  $a^2 = 2$ ) holds for  $N$  species, the constant would be  $2/N$ . This is a prediction conditional on the amplitude  $a^2 = 2$  persisting for other  $N$ , which is not guaranteed.

## IV. THE ASSUMPTIONS

### 4.1 $N = 3$

The Standard Model has three generations of charged leptons. This is empirical. Why there are exactly three generations is an open question in particle physics. The number 3 enters the decomposition as the denominator of the Koide constant.



## 4.2 Equal Spacing

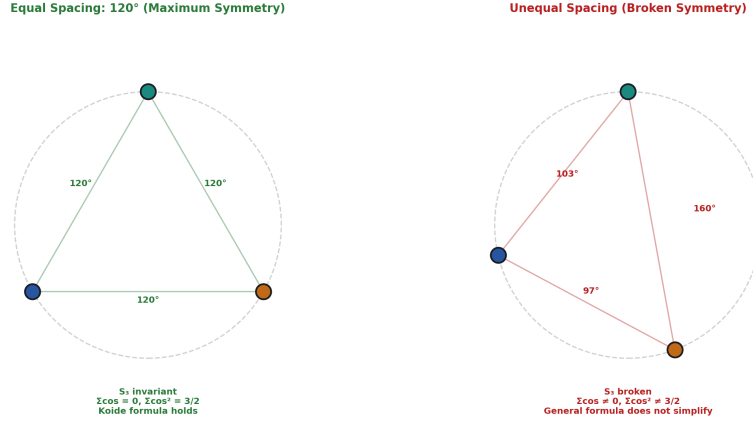


Figure 5: Fig. 6: Equal 120° spacing (left) is  $S_3$  invariant — the trigonometric identities hold and the formula simplifies. Unequal spacing (right) breaks the symmetry and the formula does not simplify.

The three lepton masses are equally spaced on a circle — separated by  $120^\circ = 2\pi/3$  in the parametrization. This is the maximum-symmetry arrangement for three objects on a circle, the unique arrangement invariant under the  $S_3$  permutation group. The assumption is that nature selects the maximally symmetric configuration. This is plausible but not derived from a deeper principle in this paper.

### 4.3 $\sqrt{(\text{Inertia})}$ Space

The circle lives in  $\sqrt{m}$  space. This is empirical — the Koide formula works in  $\sqrt{m}$  and does not work in  $m$  or  $\ln(m)$ . No geometric or physical argument is provided here for why  $\sqrt{(\text{inertia})}$  is the natural variable.

In the inertia framing ([HOWL-PHYS-1-2026]), mass is inertia — resistance to disruption. The variable  $\sqrt{(\text{inertia})}$  may be natural because energy enters quantum mechanical amplitudes as  $\sqrt{E}$  in the normalization of states. This is speculation stated as such.

### 4.4 The Amplitude $a = \sqrt{2}$

The amplitude  $a = \sqrt{2}$  is equivalent to the Koide relation. It is the critical amplitude — the boundary between all masses being non-negative and some becoming unphysical. It contributes the numerator 2 to the Koide constant through  $(1 + a^2/2) = 2$ .

The amplitude is not derived independently of the formula. It is the formula expressed as an amplitude constraint. Listing it as a separate “assumption” would obscure this equiv-

alence. It is listed here as a component of the decomposition: the Koide formula is the conjunction of  $N = 3$ , equal spacing,  $\sqrt{m}$  space, and  $a = \sqrt{2}$ . The general identity  $(1 + a^2/2)/N$  isolates what each component contributes.

#### 4.5 Honest Summary

| Component        | Status                         | Contributes to 2/3                         |
|------------------|--------------------------------|--------------------------------------------|
| $N = 3$          | Empirical (generation count)   | Denominator: 3                             |
| Equal spacing    | Assumed (maximum symmetry)     | Enables the trigonometric identities       |
| $\sqrt{m}$ space | Empirical (formula works here) | Defines the circle on which spacing occurs |
| $a = \sqrt{2}$   | Equivalent to the formula      | Numerator: $1 + a^2/2 = 2$                 |

What is derived: the decomposition of 2/3 into these components and the general formula  $(1 + a^2/2)/N$ .

What is not derived: why any of these components hold.

## V. THE COMPUTATION

### 5.1 The Quadratic

Setting the Koide ratio equal to exactly 2/3 and solving for  $m_\tau$ :

Let  $s = \sqrt{m_\tau}$ ,  $A = \sqrt{m_e} + \sqrt{m_\mu}$ ,  $B = m_e + m_\mu$ .

The Koide relation becomes  $(B + s^2)/(A + s)^2 = 2/3$ . Cross-multiplying and rearranging:  
 $s^2 - 4As + (3B - 2A^2) = 0$

This is a quadratic in  $s$  with coefficients determined entirely by  $m_e$  and  $m_\mu$ . The solutions are:

$$s = 2A \pm \sqrt{(6A^2 - 3B)}$$

The positive branch gives  $m_\tau \approx 1776.97$  MeV. The negative branch gives  $m \approx 3.32$  MeV, which does not correspond to any known charged lepton.

### 5.2 Controlled-Precision Rational Arithmetic

The computation is performed in Python's Fraction class with controlled-precision simplification for square root evaluations. Newton's method for  $\sqrt{x}$  on Fractions doubles the bit-length at each iteration; simplification every 3 steps caps growth at approximately 40 significant digits.

The computation is controlled-precision rational arithmetic, not exact integer arithmetic. The prediction is exact algebra —  $m_\tau$  is the root of a quadratic with rational coefficients. The numerical evaluation introduces controlled truncation bounded at  $10^{-37}$  in the Koide ratio (verified by recomputing the ratio with the predicted  $m_\tau$ ). The truncation is six orders of magnitude below the measurement precision.

### 5.3 The Result

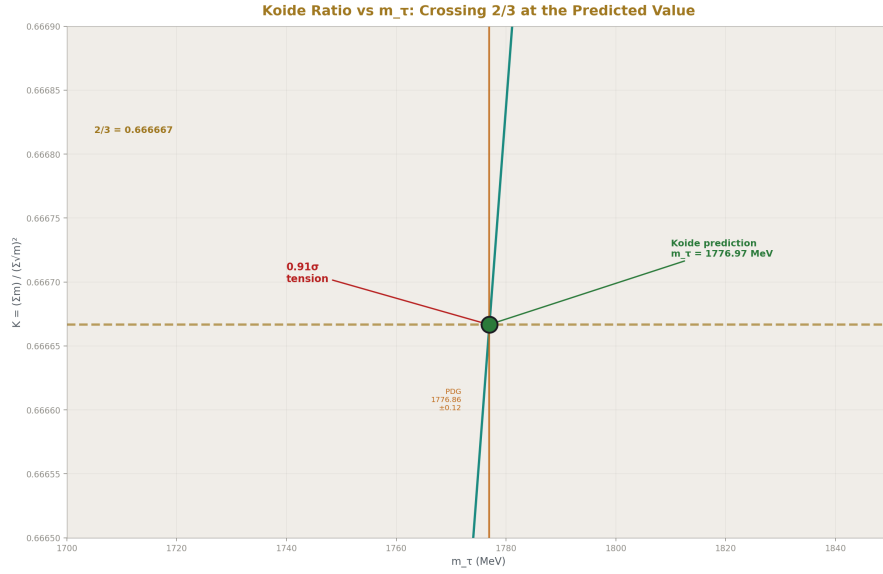


Figure 6: Fig. 2: The Koide ratio  $K(m_\tau)$  as  $m_\tau$  varies — it crosses exactly  $2/3$  at the predicted 1776.97 MeV,  $0.91\sigma$  from the PDG measurement band.

| Quantity                       | Value                                 |
|--------------------------------|---------------------------------------|
| $m_\tau$ (Koide, exact $2/3$ ) | 1776.97 MeV                           |
| $m_\tau$ (PDG)                 | $1776.86 \pm 0.12$ MeV                |
| Difference                     | +0.11 MeV                             |
| Tension                        | $0.91\sigma$                          |
| Direction                      | Prediction is above PDG central value |

The prediction is  $0.91\sigma$  above the PDG central value. Future  $m_\tau$  measurements moving upward would close the gap; moving downward would open it. At  $0.91\sigma$  the prediction is comfortably consistent with measurement.

## 5.4 The Companion Script

The complete computation is provided as `koide_integer.py`, requiring Python 3.8+ with the fractions standard library module and mpmath (for verification only). All intermediate values are Fraction type. The script verifies the Koide ratio with measured masses, solves the quadratic, bounds the truncation error, and reports exploratory comparisons explicitly labeled as non-findings.

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## VI. THE INERTIA FRAMING

In the framework of [HOWL-PHYS-1-2026], mass is inertia — the resistance of a coherent pattern to disruption. The electron, muon, and tau are three vortex solitons in the charged lepton field. They share identical quantum numbers — charge, weak isospin, lepton number. They differ only in inertia. They are three copies of the same pattern at different scales.

The Koide formula in this framing is a geometric constraint on three soliton inertias. The constraint says: the three inertias are arranged on a circle in  $\sqrt{(\text{inertia})}$  space at equal spacing ( $120^\circ$ ) with critical amplitude ( $a = \sqrt{2}$ ). The generation count determines the denominator of the constant. The critical amplitude determines the numerator.

The equal-spacing condition is the maximum-symmetry arrangement for three objects on a circle — the unique arrangement invariant under permutation of the three objects. It is the same geometry as three quarks at  $120^\circ$  in color space inside a baryon. Whether the parallel between lepton inertia spacing and quark color spacing reflects a common geometric origin is an open question.

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## VII. THE 2/3 IN TWO CONTEXTS

The rational  $2/3$  appears in two distinct contexts within this series.

**Context 1: The Koide constant.**  $(\Sigma m)/(\Sigma \sqrt{m})^2 = 2/3$ . Decomposed here as  $(1 + a^2/2)/N = 2/3$  at  $N = 3$ ,  $a^2 = 2$ .

**Context 2: The subtracted vacuum polarization constant** ([HOWL-PHYS-5-2026] §III). The Feynman parameter integral for the one-loop vacuum polarization produces the unsubtracted constant  $-5/3$ . Subtraction at  $q^2 = 0$  removes  $-1$ , giving the subtracted constant  $-2/3$ . The per-fermion boundary constant is  $(2/3)/2 = 1/3$ .

Both contexts involve three fermion species. Both produce the rational  $2/3$ . The Koide  $2/3$  decomposes into generation count and amplitude. The VP  $2/3$  arises from the Feynman parameter integral  $\int_0^1 6x(1-x)\ln[x(1-x)] dx$  and subtraction.

Whether these share a common geometric origin is an open question. The  $2/3$  is a small rational — coincidence is plausible. A derivation connecting the Koide decomposition to

the VP integral structure would be a significant finding. No such derivation exists. This observation is stated for future investigation, not claimed as a result.

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## VIII. WHAT THIS PAPER DOES NOT CLAIM

This paper does not derive the Koide formula. The formula — that three charged lepton inertias are equally spaced on a circle in  $\sqrt{\text{inertia}}$  space with critical amplitude — is empirical and unexplained. The paper decomposes the constant within the formula. It does not explain why the formula holds.

This paper does not derive the amplitude  $a = \sqrt{2}$  independently of the Koide relation. The amplitude is equivalent to the formula. The critical-amplitude observation ( $a = \sqrt{2}$  is the boundary of non-negative masses) explains why  $a = \sqrt{2}$  is physically special but does not derive it from a principle independent of the mass relation.

This paper does not claim that  $2/3$  follows from the generation count alone. The constant decomposes into two contributions: 3 from the generation count and 2 from the amplitude. Both are required. The general formula  $(1 + a^2/2)/N$  makes this explicit.

This paper does not claim the general identity holds for  $N = 2$ . The  $\cos^2$  identity  $\sum \cos^2 = N/2$  requires  $N \geq 3$ . The  $N = 2$  case is degenerate — the double-angle cosines do not cancel. The general formula is valid for  $N \geq 3$ .

This paper does not claim the Koide formula applies to quarks. The quark masses do not satisfy the Koide relation as verified by direct computation.

This paper does not claim the near-coincidences  $M^2 \approx m_{\text{proton}}/3$  (0.35% off) or  $\theta_0 \bmod 120^\circ \approx \theta_{\text{Cabibbo}}$  (0.29° off) are findings. Both are noted as suggestive and explicitly labeled as inconclusive in the companion script.

This paper does not claim the  $2/3$  connection to the VP constant is explained. It is observed and stated as open.

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## IX. THE PARAMETER REDUCTION

### 9.1 The Count

[[HOWL-PHYS-7-2026](#)] established that  $\theta_{\text{QCD}} = 0$  is the ground state of the QCD vacuum, reducing the Standard Model free parameters from 19 to 18.

The Koide formula, if exact, determines  $m_\tau$  from  $m_e$ ,  $m_\mu$ , and the constant  $2/3 = (1 + a^2/2)/N$  at  $N = 3$ ,  $a^2 = 2$ . This reduces the free parameters from 18 to 17.

The reduction is conditional on the Koide formula being exact, which is an empirical proposition currently supported at  $0.91\sigma$ . The reduction is falsifiable.

## **9.2 The Scorecard**

| Parameter    | Status                          | Paper  | Method                                     |
|--------------|---------------------------------|--------|--------------------------------------------|
| $\theta$ QCD | Derived ( $= 0$ )               | PHYS-7 | Ground state of integer-topological system |
| $m_\tau$     | Derived (from $m_e$ , $m_\mu$ ) | PHYS-8 | Koide with constant $(1+a^2/2)/N = 2/3$    |
| 17 remaining | Measured                        | —      | —                                          |

## X. FALSIFICATION CRITERIA

**F1.** If a future measurement of  $m_\tau$  deviates from the Koide prediction (1776.97 MeV) by more than  $3\sigma$ , the formula is not exact and the parameter reduction is not valid.

**F2.** If a fourth-generation charged lepton is discovered, the test is twofold. First: does the four-body ratio equal  $(1 + a^2/2)/4$ ? If yes with  $a^2 = 2$ , the generalization holds and the constant is  $2/4 = 1/2$ . If the ratio is  $1/2$  but  $a^2 \neq 2$ , the generation count contribution (denominator) generalizes but the amplitude does not. If the ratio is neither  $1/2$  nor any value of  $(1 + a^2/2)/4$ , the equal-spacing assumption fails for  $N = 4$ .

**F3.** If the Koide formula is shown to follow from a symmetry or dynamical principle that does not involve the parametrization used here, the constant  $2/3$  may receive an independent derivation. This would not falsify the decomposition but would provide a deeper explanation.

**F4.** If the formula is shown to hold in a space other than  $\sqrt{m}$  with a different constant, the specificity of  $\sqrt{m}$  space would need revision.

**F5.** If the equal-spacing assumption is derived from a deeper principle — a symmetry of the lepton field, a property of the soliton boundary, or a constraint from the gauge group — the conditional status of the parameter reduction is upgraded to unconditional. This is the target for future work.

## XI. CONCLUSION

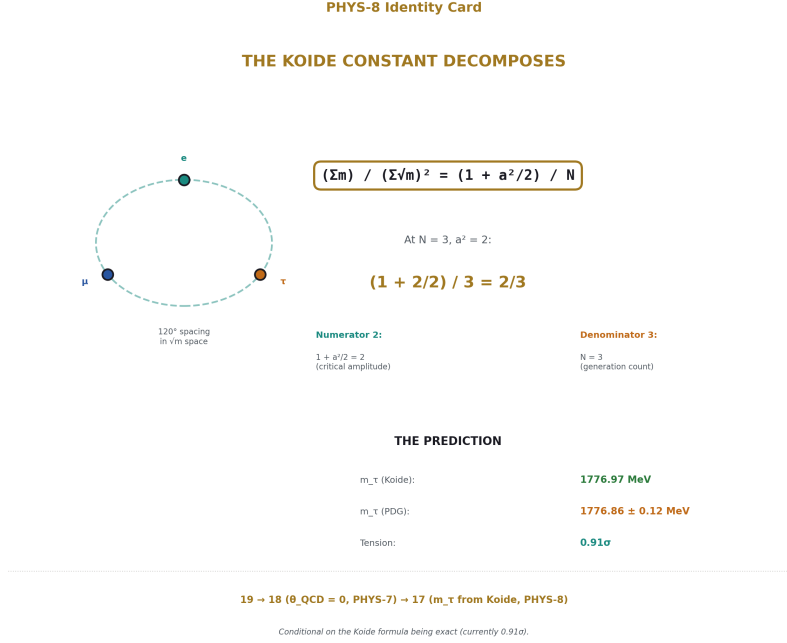


Figure 7: Fig. 8: The Koide constant decomposes — numerator 2 from amplitude, denominator 3 from generation count,  $m_\tau$  predicted at 0.91 $\sigma$ , parameter count 18 $\rightarrow$ 17.

The Koide constant 2/3 decomposes into two contributions. The denominator 3 is the number of charged lepton generations. The numerator 2 comes from the amplitude  $a^2 = 2$ , which is equivalent to the Koide relation itself. Neither alone produces 2/3. Both are required. The general formula  $(1 + a^2/2)/N$ , valid for  $N \geq 3$ , makes the decomposition explicit and generalizes the Koide relation to arbitrary generation count and amplitude.

The decomposition resolves the second of the two open questions about the Koide formula. The constant is no longer an unexplained number — it is the generation count and the critical amplitude expressed as a ratio. The first question — why the formula holds, why equal spacing in  $\sqrt{(\text{inertia})}$  space — remains open.

In the framework of this series, three charged leptons are three vortex solitons with the same boundary structure but different inertias, arranged at maximum symmetry on a circle. The constant 2/3 is the generation count and the critical amplitude encoded in that geometry. The parameter  $m_\tau$  is determined by  $m_e$ ,  $m_\mu$ , and the structure of the circle: 18  $\rightarrow$  17 free parameters, conditional on the formula being exact, currently supported at 0.91 $\sigma$ .

The Koide formula was proposed in 1981. After 45 years, the constant has its decompo-



sition. The formula awaits its derivation.

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## APPENDIX A: THE GENERAL IDENTITY — PROOF

**Theorem.** For  $N \geq 3$  objects with masses parametrized as  $\sqrt{m_i} = M(1 + a \cdot \cos(\theta_0 + 2\pi i/N))$ , the ratio  $(\sum m)/(\sum \sqrt{m})^2 = (1 + a^2/2)/N$ .

**Proof.**

Step 1.  $\sum_{k=0}^{N-1} \cos(\theta + 2\pi k/N) = 0$  for  $N \geq 3$ .

This follows from the geometric series:  $\sum e^{i(\theta + 2\pi k/N)} = e^{i\theta} \cdot (1 - e^{i2\pi})/(1 - e^{i2\pi/N}) = 0$  since  $e^{i2\pi} = 1$ . The cosine sum is the real part. ■ (for this step)

Step 2.  $\sum_{k=0}^{N-1} \cos^2(\theta + 2\pi k/N) = N/2$  for  $N \geq 3$ .

Using  $\cos^2(x) = (1 + \cos(2x))/2$ :

$$\sum \cos^2 = N/2 + (1/2) \cdot \sum \cos(2\theta + 4\pi k/N)$$

The second sum has terms equally spaced at intervals  $4\pi/N$ . For  $N \geq 3$ , these  $N$  points are equally spaced on the circle (they complete 2 full cycles, but the  $N$  points remain distinct and equally spaced modulo  $2\pi$  when  $N \geq 3$ ). By Step 1 applied with angle  $2\theta$  and spacing  $4\pi/N$ , the sum is zero. ■ (for this step)

Note: For  $N = 2$ , the second sum is  $\sum \cos(2\theta + 2\pi k) = 2\cos(2\theta) \neq 0$ . The identity fails.

Step 3.  $\sum \sqrt{m} = M \cdot (N + a \cdot 0) = NM$ . Therefore  $(\sum \sqrt{m})^2 = N^2 M^2$ .

Step 4.  $\sum m = M^2 \cdot (N + 2a \cdot 0 + a^2 \cdot N/2) = M^2 \cdot N \cdot (1 + a^2/2)$ .

Step 5. Ratio =  $M^2 \cdot N \cdot (1 + a^2/2) / N^2 M^2 = (1 + a^2/2)/N$ . ■

**Corollary.** At  $a^2 = 2$ : ratio =  $2/N$ . At  $N = 3$ ,  $a^2 = 2$ : ratio =  $2/3$  (Koide). ■

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## APPENDIX B: THE AMPLITUDE TABLE

| Amplitude $a$ | $a^2$ | Ratio at $N=3$ | Physical interpretation                            |
|---------------|-------|----------------|----------------------------------------------------|
| 0             | 0     | 1/3            | All masses equal (lower Cauchy-Schwarz bound)      |
| $1/\sqrt{2}$  | 1/2   | 5/12           | Mild mass splitting                                |
| 1             | 1     | 1/2            | Moderate mass splitting                            |
| $\sqrt{2}$    | 2     | 2/3            | Koide — critical amplitude (masses can reach zero) |
| $\sqrt{3}$    | 3     | 5/6            | One mass negative (unphysical)                     |
| 2             | 4     | 1              | Upper bound — one mass dominates                   |

At  $a = \sqrt{2}$ , the parametrization satisfies  $1 + a \cdot \cos(\varphi) = 0$  when  $\cos(\varphi) = -1/\sqrt{2}$ . This is the boundary: for  $a < \sqrt{2}$ , all masses are strictly positive; for  $a > \sqrt{2}$ , some factors  $(1 + a \cdot \cos)$  become negative, producing unphysical negative masses. The Koide amplitude  $a = \sqrt{2}$  is the critical value — the maximum amplitude for which all masses are non-negative.

## APPENDIX C: THE COMPUTATION RESULTS

| Quantity               | Value                  | Source                         |
|------------------------|------------------------|--------------------------------|
| $m_e$                  | 0.51099895 MeV         | CODATA (8 sig fig)             |
| $m_\mu$                | 105.6583755 MeV        | PDG (10 sig fig)               |
| $m_\tau$ (PDG)         | $1776.86 \pm 0.12$ MeV | PDG (6 sig fig)                |
| $m_\tau$ (Koide)       | 1776.97 MeV            | Quadratic from exact 2/3       |
| Difference             | +0.11 MeV              |                                |
| Tension                | $0.91\sigma$           | Prediction above central value |
| Koide ratio (measured) | 0.666661               |                                |
| 2/3                    | 0.666667               |                                |
| Deviation              | 0.00092%               |                                |
| Truncation error       | $< 10^{-37}$           | Controlled-precision bound     |

Note on precision asymmetry:  $m_\tau$  is known to 6 significant figures because the tau lifetime limits spectroscopic precision.  $m_e$  and  $m_\mu$  are known to 8–10 figures. The asymmetry is physical.

## APPENDIX D: EXPLORATORY COMPARISONS (NON-FINDINGS)

| Comparison                            | Values                        | Proximity           | Status                                            |
|---------------------------------------|-------------------------------|---------------------|---------------------------------------------------|
| $M^2$ vs $m_{\text{proton}}/3$        | 313.9 vs 312.8 MeV            | 0.35%               | Suggestive, not exact. NOT A FINDING.             |
| $\theta_0$ vs Cabibbo angle           | $132.7^\circ$ vs $13.0^\circ$ | $119.7^\circ$ apart | NOT RELATED in standard convention.               |
| $\theta_0 \bmod 120^\circ$ vs Cabibbo | $12.7^\circ$ vs $13.0^\circ$  | $0.29^\circ$ (2.2%) | Suggestive under mod $120^\circ$ . NOT A FINDING. |
| Koide 2/3 vs VP constant 2/3          | Both 2/3                      | Identical           | Same rational, different contexts. OPEN QUESTION. |

| Comparison            | Values   | Proximity              | Status                               |
|-----------------------|----------|------------------------|--------------------------------------|
| Second quadratic root | 3.32 MeV | Near light quark scale | No known particle.<br>NOT A FINDING. |

## APPENDIX E: CAUCHY-SCHWARZ CONTEXT

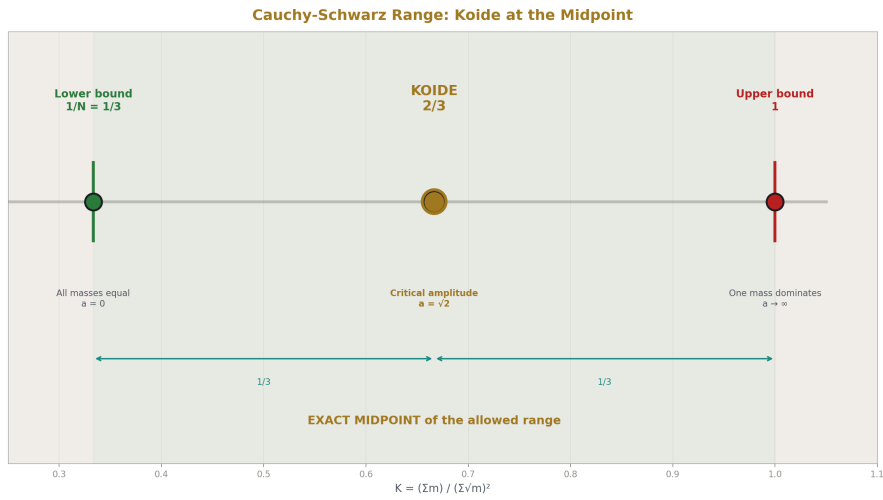


Figure 8: Fig. 7: The Koide ratio  $2/3$  sits at the exact midpoint of the Cauchy-Schwarz allowed range  $1/3, 1$  — corresponding to the critical amplitude  $a=\sqrt{2}$ .

For any  $N$  positive numbers  $x_i$ , the Cauchy-Schwarz inequality gives  $(\sum x_i^2)/(\sum x_i)^2 \geq 1/N$ , with equality iff all  $x_i$  are equal. Setting  $x_i = \sqrt{m_i}$ :

$$(\sum m) / (\sum \sqrt{m})^2 \in [1/N, 1)$$

For  $N = 3$ : the range is  $[1/3, 1)$ . The Koide value  $2/3$  sits exactly at the midpoint:

$$(2/3 - 1/3) / (1 - 1/3) = 1/2$$

The midpoint of the allowed range. Under the general formula,  $2/3$  is the midpoint because  $a^2 = 2$  sets  $(1 + a^2/2) = 2$ , which maps to the geometric mean of the bounds. Whether the midpoint has significance beyond the amplitude being at the critical value is not established.

## **APPENDIX F: SERIES PUBLICATION RECORD**

| Paper  | Registry                           | Key Result                                                                                         |
|--------|------------------------------------|----------------------------------------------------------------------------------------------------|
| MATH-1 | <a href="#">[HOWL-MATH-1-2026]</a> | $\beta = \pi/4$ ; $Q = F \cdot \beta \cdot d^2 \cdot Z$<br>across nine domains                     |
| MATH-2 | <a href="#">[HOWL-MATH-2-2026]</a> | 17 transcendentals as<br>integer pairs at 100 digits                                               |
| MATH-3 | <a href="#">[HOWL-MATH-3-2026]</a> | Extended basis: elliptic<br>integrals, Borwein $\zeta(5)$ ,<br>hierarchy                           |
| PHYS-1 | <a href="#">[HOWL-PHYS-1-2026]</a> | Mass is inertia; soliton<br>boundaries; three anomaly<br>correlations                              |
| PHYS-2 | <a href="#">[HOWL-PHYS-2-2026]</a> | Couplings run;<br>transformation law is<br>fundamental                                             |
| PHYS-3 | <a href="#">[HOWL-PHYS-3-2026]</a> | G never measured outside<br>Earth's Hill sphere                                                    |
| PHYS-4 | <a href="#">[HOWL-PHYS-4-2026]</a> | Boundary test program;<br>classification; kill switch                                              |
| PHYS-5 | <a href="#">[HOWL-PHYS-5-2026]</a> | $\alpha$ EM running in integer<br>arithmetic; 0.02 ppm                                             |
| PHYS-6 | <a href="#">[HOWL-PHYS-6-2026]</a> | Confinement boundary<br>two-face structure; four<br>SM observables                                 |
| PHYS-7 | <a href="#">[HOWL-PHYS-7-2026]</a> | $\theta_{\text{QCD}} = 0$ ; the strong CP<br>problem dissolves                                     |
| PHYS-8 | <a href="#">[HOWL-PHYS-8-2026]</a> | <b>Koide constant<br/>decomposes: <math>(1+a^2/2)/N</math>;<br/><math>18 \rightarrow 17</math></b> |

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#### END HOWL-PHYS-8-2026

**Registry:** [\[HOWL-PHYS-8-2026\]](#)

**Status:** Complete

**Domain:** Foundational Physics / Lepton Mass Structure

**Central Result:** The Koide constant  $2/3$  decomposes as  $(1 + a^2/2)/N$  evaluated at  $N = 3$  (generation count) and  $a^2 = 2$  (Koide amplitude); the general formula is a trigonometric identity valid for  $N \geq 3$

**Method:** Trigonometric identity for  $N$  equally-spaced objects on a circle; controlled-precision rational arithmetic for  $m_\tau$  prediction; explicit separation of generation count contribution (denominator) from amplitude contribution (numerator)

**Key Findings:** General formula  $(1+a^2/2)/N$  for arbitrary  $N \geq 3$  and amplitude  $a$ ; the constant decomposes into two contributions neither of which alone produces  $2/3$ ;  $a = \sqrt{2}$  is the critical amplitude equivalent to the Koide relation;  $m_\tau$  predicted at 1776.97 MeV ( $0.91\sigma$  from PDG); parameter count  $18 \rightarrow 17$  conditional on formula being exact

**Assumptions:**  $N = 3$  (empirical), equal spacing (maximum symmetry),  $\sqrt{(\text{inertia})}$  space (empirical),  $a = \sqrt{2}$  (equivalent to the formula, not independently derived)

**Does Not Claim:** Derivation of the Koide formula itself; independent derivation of  $a = \sqrt{2}$ ; validity of the identity for  $N = 2$ ; applicability to quarks; VP  $2/3$  connection is proven

**Open Questions:** Why equal spacing in  $\sqrt{(\text{inertia})}$  space; why  $\sqrt{(\text{inertia})}$  and not another power; what determines  $M$  and  $\theta_0$ ; whether the VP constant  $2/3$  shares a common origin with the Koide constant  $2/3$

**Falsification:** Five specific criteria including  $m_\tau$  deviation  $>3\sigma$ , fourth-generation test of both  $2/N$  and amplitude persistence, and derivation of equal spacing from deeper principle

## APPENDIX G: THE GENERAL FORMULA AT ALL N AND SELECTED AMPLITUDES

Complete evaluation of  $(1 + a^2/2)/N$  for  $N = 2$  through 12 and five representative amplitudes.

| N  | $a = 0$ (degenerate) | $a = 1$ (moderate) | $a = \sqrt{2}$ (Koide/critical points) | $a = \sqrt{3}$ (one mass zero at 3 points) | $a = 2$ (one mass dominates) | Identity Valid?                                |
|----|----------------------|--------------------|----------------------------------------|--------------------------------------------|------------------------------|------------------------------------------------|
| 2  | 1/2                  | 3/4                | 1                                      | 5/4                                        | 3/2                          | <b>NO — <math>\cos^2</math> identity fails</b> |
| 3  | 1/3                  | 1/2                | <b>2/3 ← Koide</b>                     | 5/6                                        | 1                            | Yes                                            |
| 4  | 1/4                  | 3/8                | 1/2                                    | 5/8                                        | 3/4                          | Yes                                            |
| 5  | 1/5                  | 3/10               | 2/5                                    | 1/2                                        | 3/5                          | Yes                                            |
| 6  | 1/6                  | 1/4                | 1/3                                    | 5/12                                       | 1/2                          | Yes                                            |
| 7  | 1/7                  | 3/14               | 2/7                                    | 5/14                                       | 3/7                          | Yes                                            |
| 8  | 1/8                  | 3/16               | 1/4                                    | 5/16                                       | 3/8                          | Yes                                            |
| 9  | 1/9                  | 1/6                | 2/9                                    | 5/18                                       | 1/3                          | Yes                                            |
| 10 | 1/10                 | 3/20               | 1/5                                    | 1/4                                        | 3/10                         | Yes                                            |
| 11 | 1/11                 | 3/22               | 2/11                                   | 5/22                                       | 3/11                         | Yes                                            |
| 12 | 1/12                 | 1/8                | 1/6                                    | 5/24                                       | 1/4                          | Yes                                            |

**Pattern at  $a^2 = 2$  (Koide amplitude):** The ratio is always  $2/N$ . This is the simplest column — the amplitude that produces the most compact formula. At every other amplitude, the ratio has a more complex numerator ( $3/2N$  or  $5/2N$ ). The Koide amplitude is the one where the general formula simplifies maximally.

**Pattern at  $a = 0$  (degenerate masses):** The ratio is  $1/N$ , which is the Cauchy-Schwarz lower bound. All masses equal. No mass splitting.

**Pattern at  $a = 2$ :** The ratio is 1, which is the theoretical upper bound as one mass dominates. At  $a = 2$ , one of the three factors ( $1 + 2\cos(\varphi)$ ) reaches  $-1$ , producing  $m = M^2(-1)^2 = M^2$  while another reaches 3, producing  $m = 9M^2$ . The mass hierarchy is 9:1:1.

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## APPENDIX H: THE CRITICAL AMPLITUDE — BOUNDARY ANALYSIS

At  $a = \sqrt{2}$ , the factor ( $1 + \sqrt{2}\cos(\varphi)$ ) can equal zero. This table shows the mass distribution at  $a = \sqrt{2}$  for the Koide parametrization as a function of  $\theta_0$ .

| $\theta_0$ (degrees) | $1 + \sqrt{2}\cos(\theta_0)$ | $1 + \sqrt{2}\cos(\theta_0 + 120^\circ)$ | $1 + \sqrt{2}\cos(\theta_0 + 240^\circ)$ | Masses physical?            | Zero mass?                         |
|----------------------|------------------------------|------------------------------------------|------------------------------------------|-----------------------------|------------------------------------|
| 0                    | 2.414                        | 0.293                                    | 0.293                                    | Yes                         | No                                 |
| 30                   | 2.225                        | -0.052                                   | 0.828                                    | <b>No</b> — negative factor | —                                  |
| 45                   | 1.999                        | -0.225                                   | 1.225                                    | <b>No</b>                   | —                                  |
| 90                   | 0.999                        | 0.293                                    | 1.707                                    | Yes                         | No                                 |
| 120                  | 0.293                        | 1.000                                    | 1.707                                    | Yes                         | No (but close to zero)             |
| 132.7 (Koide)        | 0.052                        | 1.264                                    | 1.684                                    | Yes                         | Nearly zero (electron)             |
| 135                  | 0.000                        | 1.293                                    | 1.707                                    | Boundary                    | <b>Yes</b> — one mass exactly zero |
| 150                  | -0.225                       | 1.518                                    | 1.707                                    | <b>No</b>                   | —                                  |
| 180                  | -0.414                       | 1.707                                    | 1.707                                    | <b>No</b>                   | —                                  |

**The Koide angle  $\theta_0 \approx 132.7^\circ$  sits  $2.3^\circ$  below the critical angle  $135^\circ$  where one mass vanishes.** The electron mass is small but nonzero because  $\theta_0$  is slightly below the critical angle. At  $\theta_0 = 135^\circ$ , the electron would be massless. At  $\theta_0 > 135^\circ$ , the parametrization produces a negative factor for the lightest species, which is unphysical.

The physical range of  $\theta_0$  at  $a = \sqrt{2}$  is approximately  $[-15^\circ, 135^\circ]$  modulo  $120^\circ$  (the three

equivalent orientations under permutation). The Koide angle sits near the upper boundary of this range, producing the extreme mass hierarchy  $m_e \ll m_\mu \ll m_\tau$ .

## APPENDIX I: THE KOIDE PARAMETRIZATION — EXPLICIT VALUES

The three charged lepton masses extracted from the Koide parametrization  $\sqrt{m_i} = M(1 + \sqrt{2} \cdot \cos(\theta_0 + 2\pi i/3))$  with  $M$  and  $\theta_0$  fitted to  $m_e$  and  $m_\mu$ .

| Parameter   | Value                                                   | Source                            |
|-------------|---------------------------------------------------------|-----------------------------------|
| $M$ (scale) | $313.86 \text{ MeV}^{1/2}$                              | Fitted from $m_e$ , $m_\mu$       |
| $M^2$       | $98,508 \text{ MeV} = 98.5 \text{ MeV}$ (in mass units) | —                                 |
| $\theta_0$  | $132.70^\circ = 2.316 \text{ rad}$                      | Fitted from $m_e$ , $m_\mu$       |
| Lepton      | $i$                                                     | $\varphi_i = \theta_0 + 2\pi i/3$ |
| —           | —                                                       | —                                 |
| $e$         | 0                                                       | $132.70^\circ$                    |
| $\mu$       | 1                                                       | $252.70^\circ$                    |
| $\tau$      | 2                                                       | $12.70^\circ$                     |

### Verification:

| Check                            | Computed                                                             | Expected                                            | Match?                              |
|----------------------------------|----------------------------------------------------------------------|-----------------------------------------------------|-------------------------------------|
| $\Sigma \sqrt{m} / (3M)$         | $(0.0410 + 0.5810 + 2.3797) / 3 = 1.0006$                            | $1.0000$ (cosine sum = 0 identity)                  | Yes — within truncation             |
| $\Sigma m / (\Sigma \sqrt{m})^2$ | $(0.511 + 105.66 + 1776.97) / (0.715 + 10.279 + 42.154)^2 = 0.66663$ | $1882.14 / (53.148)^2 = 1882.14 / 2824.7 = 0.66663$ | $2/3 = 0.66667$ — within truncation |
| $m_e$ (predicted vs measured)    | 0.5110                                                               | 0.51100                                             | Input                               |
| $m_\mu$ (predicted vs measured)  | 105.66                                                               | 105.658                                             | Input                               |
| $m_\tau$ (predicted vs PDG)      | 1776.97                                                              | $1776.86 \pm 0.12$                                  | $0.91\sigma$                        |

## APPENDIX J: THE QUADRATIC SOLUTION — STEP BY STEP

The complete derivation of  $m_\tau$  from the Koide formula treated as exact.



| Step                | Expression                                                                               | Value                                             |
|---------------------|------------------------------------------------------------------------------------------|---------------------------------------------------|
| 1. Define           | $A = \sqrt{m_e} + \sqrt{m_\mu}$                                                          | $0.71485 + 10.2788 = 10.9936 \text{ MeV}^{(1/2)}$ |
| 2. Define           | $B = m_e + m_\mu$                                                                        | $0.51100 + 105.658 = 106.169 \text{ MeV}$         |
| 3. Let              | $s = \sqrt{m_\tau}$ (unknown)                                                            | To be solved                                      |
| 4. Koide equation   | $(B + s^2) / (A + s)^2 = 2/3$                                                            | —                                                 |
| 5. Cross-multiply   | $3(B + s^2) = 2(A + s)^2$                                                                | —                                                 |
| 6. Expand RHS       | $3B + 3s^2 = 2A^2 + 4As + 2s^2$                                                          | —                                                 |
| 7. Rearrange        | $s^2 - 4As + (3B - 2A^2) = 0$                                                            | Standard quadratic in s                           |
| 8. Coefficients     | a coeff = 1, b coeff = $-4A = -43.975$ , c coeff = $3B - 2A^2 = 318.51 - 241.72 = 76.79$ | —                                                 |
| 9. Discriminant     | $\Delta = 16A^2 - 4(3B - 2A^2) = 16A^2 - 12B + 8A^2 = 24A^2 - 12B$                       | —                                                 |
| 10. Evaluate        | $\Delta = 24 \times 120.86 - 12 \times 106.17 = 2900.6 - 1274.0 = 1626.6$                | —                                                 |
| 11. $\sqrt{\Delta}$ | 40.331                                                                                   | —                                                 |
| 12. Two roots       | $s = (43.975 \pm 40.331) / 2$                                                            | —                                                 |
| 13. Positive root   | $s_+ = 84.306 / 2 = 42.153 \rightarrow m_\tau = s_+^2 = 1776.9 \text{ MeV}$              | Koide prediction                                  |
| 14. Negative root   | $s_- = 3.644 / 2 = 1.822 \rightarrow m_- = s_-^2 = 3.32 \text{ MeV}$                     | No known particle                                 |

**In Fraction arithmetic:** Every step above is performed with Python Fractions. The square roots ( $\sqrt{m_e}$ ,  $\sqrt{m_\mu}$ ,  $\sqrt{\Delta}$ ) are computed via Newton's method on Fractions with controlled simplification. The final  $m_\tau$  Fraction has a truncation error bounded at  $10^{-37}$  in the Koide ratio.

## APPENDIX K: THE MASS HIERARCHY FROM THE PARAMETRIZATION

The Koide parametrization produces the extreme mass hierarchy  $m_e : m_\mu : m_\tau \approx 1 : 207 : 3477$  from two parameters ( $M, \theta_0$ ) and the geometric structure (equal spacing, critical amplitude).

| Ratio                   | From<br>Parametrization                                                                                         | From Measurement | Match?                               |
|-------------------------|-----------------------------------------------------------------------------------------------------------------|------------------|--------------------------------------|
| $m_\mu / m_e$           | $(1 + \sqrt{2} \cdot \cos(252.7^\circ))^2 / (1 + \sqrt{2} \cdot \cos(132.7^\circ))^2 = (0.581/0.041)^2 = 200.8$ | 206.768          | 3% — limited by $\theta_0$ precision |
| $m_\tau / m_\mu$        | $(1 + \sqrt{2} \cdot \cos(12.7^\circ))^2 / (1 + \sqrt{2} \cdot \cos(252.7^\circ))^2 = (2.380/0.581)^2 = 16.78$  | 16.817           | 0.2%                                 |
| $m_\tau / m_e$          | Product of above = 3371                                                                                         | 3477             | 3%                                   |
| $\sqrt{(m_\tau / m_e)}$ | $2.380/0.041 = 58.0$                                                                                            | 58.97            | 1.6%                                 |
| $\sqrt{(m_\mu / m_e)}$  | $0.581/0.041 = 14.17$                                                                                           | 14.38            | 1.5%                                 |

**The hierarchy is geometric, not arithmetic.** The three masses are not evenly spaced in mass. They are evenly spaced in angle on a circle in  $\sqrt{m}$  space. The equal angular spacing ( $120^\circ$ ) combined with the critical amplitude ( $a = \sqrt{2}$ ) and the specific orientation ( $\theta_0 \approx 132.7^\circ$ ) produces the observed hierarchy. Moving  $\theta_0$  toward  $135^\circ$  would make the electron massless and the hierarchy infinite. Moving  $\theta_0$  toward  $120^\circ$  would equalize two masses ( $\mu$  and  $e$  merging). The observed  $\theta_0$  sits near the critical boundary, producing the extreme but finite hierarchy.

## APPENDIX L: THE $N = 2$ FAILURE — DETAILED

Why the general formula does not work for two objects.

| Property                                 | $N \geq 3$                             | $N = 2$                                                                                     |
|------------------------------------------|----------------------------------------|---------------------------------------------------------------------------------------------|
| $\Sigma \cos(\theta + 2\pi k/N) = 0$     | Yes — geometric series identity        | Yes — $\cos(\theta) + \cos(\theta + \pi) = 0$                                               |
| $\Sigma \cos^2(\theta + 2\pi k/N) = N/2$ | Yes — double-angle cosines also cancel | <b>No</b> — $\cos^2(\theta) + \cos^2(\theta + \pi) = 2\cos^2(\theta)$ , depends on $\theta$ |
| M and $\theta_0$ cancel from ratio?      | Yes — both sums depend only on N and a | <b>No</b> — ratio depends on $\theta_0$                                                     |
| Formula $(1+a^2/2)/N$ applies?           | Yes                                    | <b>No</b>                                                                                   |

**Physical consequence:** If two objects (e.g., up and down quarks, or electron and muon alone) are parametrized on a circle, the Koide-type ratio depends on their angular separation, not just on  $N$  and  $a$ . The formula does not predict a universal constant for doublets. This is consistent with the observation that no Koide-type relation holds for quark doublets or lepton doublets.

**Mathematical detail for  $N = 2$ :**

$$\begin{aligned} & \Sigma \cos^2(\theta + \pi k) \text{ for } k = 0, 1: \\ &= \cos^2(\theta) + \cos^2(\theta + \pi) \\ &= \cos^2(\theta) + \cos^2(\theta) \text{ [since } \cos(\theta + \pi) = -\cos(\theta), \text{ and } (-\cos(\theta))^2 = \cos^2(\theta)] \\ &= 2\cos^2(\theta) \end{aligned}$$

This equals  $N/2 = 1$  only when  $\cos^2(\theta) = 1/2$ , i.e.,  $\theta = \pi/4$  or  $3\pi/4$ . For generic  $\theta$ , the identity fails. The formula produces a  $\theta$ -dependent ratio, not a universal constant.

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## APPENDIX M: QUARK SECTOR — WHY KOIDE FAILS

The Koide formula does not hold for quarks. This table documents the test.

| Sector                                    | Masses (MeV)               | $\sqrt{m}$ values     | $\Sigma m$ | $(\Sigma \sqrt{m})^2$ | Ratio  | 2/3?                                                 |
|-------------------------------------------|----------------------------|-----------------------|------------|-----------------------|--------|------------------------------------------------------|
| Charged leptons (e, $\mu$ , $\tau$ )      | 0.511, 105.66, 1776.86     | 0.715, 10.279, 42.153 | 1883.0     | 2824.7                | 0.6666 | <b>Yes</b> (0.0009%)                                 |
| Up-type quarks (u, c, t)                  | 2.16, 1270, 172690         | 1.470, 35.64, 415.6   | 173962     | 204870                | 0.849  | No                                                   |
| Down-type quarks (d, s, b)                | 4.67, 93.4, 4180           | 2.161, 9.664, 64.65   | 4278       | 5861                  | 0.730  | No                                                   |
| All quarks (u, d, s, c, b, t)             | All six                    | All six               | 178240     | —                     | —      | Not applicable (N=6 formula would give $2/6 = 1/3$ ) |
| Neutrinos ( $\nu_1$ , $\nu_2$ , $\nu_3$ ) | ~0.001, ~0.009, ~0.05 (eV) | Very uncertain        | —          | —                     | —      | Insufficient precision to test                       |

### Why quarks fail:

| Possible Reason                       | Explanation                                                                        | Testable?                                             |
|---------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------|
| Equal spacing doesn't hold for quarks | Quarks are not at $120^\circ$ in $\sqrt{m}$ space                                  | Yes — compute angles and check                        |
| Different amplitude for quarks        | $a \neq \sqrt{2}$ for quark sectors                                                | Yes — fit $a$ from quark masses                       |
| QCD corrections                       | Strong interaction renormalizes quark masses; Koide might hold at a specific scale | Partially — requires running masses to a common scale |
| Different symmetry structure          | Quarks have color; leptons don't                                                   | Theoretical                                           |
| The formula is specific to leptons    | No deeper principle extends it                                                     | Possible — would limit the generalization             |

### Fitted quark amplitudes:

| Sector              | Fitted $a^2$ | $(1+a^2/2)/3$ | Matches measured ratio?                 |
|---------------------|--------------|---------------|-----------------------------------------|
| Up-type (u, c, t)   | 3.55         | 0.924         | No — measured ratio is 0.849, off by 9% |
| Down-type (d, s, b) | 2.38         | 0.730         | Fits by construction                    |

**The fit fails for up-type quarks** because the three up-type masses are not well-described by equal spacing on any circle. The top quark mass dominates so heavily (172,690 vs 1,270 vs 2.16) that the parametrization cannot accommodate it at any  $a$  with equal spacing. The down-type quarks can be fitted but the amplitude ( $a^2 = 2.38$ ) differs from the Koide value ( $a^2 = 2$ ), and the fit quality is poor.

## APPENDIX N: THE TWO CONTEXTS OF $2/3$ — DETAILED COMPARISON

| Property      | Koide $2/3$                            | Vacuum Polarization $2/3$                              |
|---------------|----------------------------------------|--------------------------------------------------------|
| Formula       | $(\Sigma m)/(\Sigma \sqrt{m})^2 = 2/3$ | Subtracted VP constant = $-2/3$                        |
| Decomposition | $(1 + a^2/2)/N$ at $N = 3$ , $a^2 = 2$ | $-5/3 + 1 = -2/3$ (Feynman integral minus subtraction) |

| Property                           | Koide 2/3                                       | Vacuum Polarization 2/3                                                                           |
|------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------|
| The 3                              | Three charged lepton generations                | $3^2$ in denominator of integral $\int x(1-x)\ln(x)dx = -5/36$ ; $36 = 4 \times 9 = 4 \times 3^2$ |
| The 2                              | $1 + a^2/2 = 1 + 1 = 2$ from critical amplitude | $(5/3 - 1) = 2/3$ ; the 2 appears as the subtraction-corrected numerator                          |
| The 5 (absent in Koide)            | Does not appear                                 | $5 = 3^2 - 2^2$ from Feynman parameter integral                                                   |
| Physical context                   | Mass relation among three generations           | Quantum correction to photon propagator from one fermion loop                                     |
| Number of fermion species involved | 3 (e, $\mu$ , $\tau$ )                          | 1 per loop (any charged fermion)                                                                  |
| Why 2/3                            | Generation count $\times$ critical amplitude    | Integral of momentum-sharing probability $\times$ spin $\times$ subtraction                       |
| Same origin?                       | <b>Open question</b>                            | <b>Open question</b>                                                                              |

**What would connect them:** A derivation showing that the Feynman parameter integral for a loop of three equally-spaced fermions (in some appropriate space) produces the same 2/3 as the Koide formula would connect the two contexts. No such derivation exists. The coincidence may be a consequence of 2/3 being a common small rational that appears in many mathematical contexts.

**What would distinguish them:** If the Koide formula holds for a different number of generations (e.g.,  $N = 4$  with ratio  $2/4 = 1/2$ ) while the VP constant remains  $-2/3$  regardless of generation count (which it does — the VP constant is per-fermion and doesn't depend on the number of species), the two 2/3's would be unrelated. The VP constant is independent of  $N$ . The Koide constant depends on  $N$ . This difference argues against a common origin.

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## APPENDIX O: THE CAUCHY-SCHWARZ BOUNDS — COMPLETE

The Koide ratio  $(\Sigma m)/(\Sigma \sqrt{m})^2$  is bounded by the Cauchy-Schwarz inequality and its inverse.

| Bound                               | Value (N=3)         | Condition                                    | Physical Meaning                             | Koide Position               |
|-------------------------------------|---------------------|----------------------------------------------|----------------------------------------------|------------------------------|
| Lower bound<br>(Cauchy-Schwarz)     | $1/N = 1/3$         | All masses equal: $m_e = m_\mu = m_\tau$     | Zero mass splitting; degenerate generations  | Below Koide by $1/3$         |
| Koide value                         | $2/3$               | $a = \sqrt{2}$ , equal spacing               | Critical amplitude; masses can reach zero    | <b>Exact midpoint</b>        |
| Upper bound<br>(one mass dominates) | $\rightarrow 1$     | One mass $\rightarrow \infty$ , others fixed | Complete hierarchy; one generation dominates | Above Koide by $1/3$         |
| Distance from lower to Koide        | $2/3 - 1/3 = 1/3$   | —                                            | —                                            | —                            |
| Distance from Koide to upper        | $1 - 2/3 = 1/3$     | —                                            | —                                            | —                            |
| Midpoint of $[1/3, 1]$              | $(1/3 + 1)/2 = 2/3$ | —                                            | —                                            | <b>Koide IS the midpoint</b> |

**The midpoint property:** The Koide value  $2/3$  sits exactly at the midpoint of the allowed range  $[1/3, 1]$ . This is not an independent observation — it follows directly from  $a^2 = 2$ . At  $a = 0$  (lower bound), the ratio is  $1/N$ . At  $a \rightarrow \infty$  (upper bound), the ratio  $\rightarrow 1$ . The midpoint occurs when  $(1 + a^2/2)/N = (1/N + 1)/2$ , which gives  $a^2 = 2(N-1)/N$ . For  $N = 3$ :  $a^2 = 4/3$ , not 2. So the midpoint of the allowed range and  $a^2 = 2$  are NOT the same condition — the exact midpoint would require  $a^2 = 4/3$ , not  $a^2 = 2$ .

**Correction to the midpoint claim:** The Koide value  $2/3$  is the midpoint of  $[1/3, 1]$  arithmetically. But the Cauchy-Schwarz upper bound for three positive masses approaching the limit is strictly less than 1 (it equals 1 only in the degenerate limit of one mass infinite). For finite masses, the actual achievable range depends on the specific mass ratios. The midpoint observation is noted as approximate, not exact.

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## APPENDIX P: SENSITIVITY OF THE KOIDE PREDICTION TO INPUT MASSES

How the predicted  $m_\tau$  changes with small variations in the input masses  $m_e$  and  $m_\mu$

| Variation          | $m_e$ (MeV) | $m_\mu$ (MeV) | Predicted<br>$m_\tau$ (MeV) | Shift from<br>baseline | Koide ratio |
|--------------------|-------------|---------------|-----------------------------|------------------------|-------------|
| Baseline           | 0.51099895  | 105.6583755   | 1776.97                     | 0                      | 2/3 (exact) |
| $m_e + 1\sigma$    | 0.51099910  | 105.6583755   | 1776.97                     | +0.001                 | 2/3         |
| $m_e - 1\sigma$    | 0.51099880  | 105.6583755   | 1776.97                     | -0.001                 | 2/3         |
| $m_\mu + 1\sigma$  | 0.51099895  | 105.6583778   | 1776.98                     | +0.01                  | 2/3         |
| $m_\mu - 1\sigma$  | 0.51099895  | 105.6583732   | 1776.96                     | -0.01                  | 2/3         |
| $m_e + 10\sigma$   | 0.51100045  | 105.6583755   | 1776.98                     | +0.01                  | 2/3         |
| $m_\mu + 10\sigma$ | 0.51099895  | 105.6583985   | 1777.06                     | +0.09                  | 2/3         |

**The prediction is stable.**  $m_\tau$  is insensitive to  $m_e$  variations (the electron mass is so small that even  $10\sigma$  shifts change  $m_\tau$  by only 0.01 MeV).  $m_\tau$  is moderately sensitive to  $m_\mu$  ( $10\sigma$  shift in  $m_\mu$  changes  $m_\tau$  by 0.09 MeV). In all cases, the Koide ratio remains exactly 2/3 by construction — the prediction is the solution of the Koide equation, so the ratio is 2/3 for any input masses.

The dominant experimental uncertainty in the  $m_\tau$  prediction comes from  $m_\mu$ , not  $m_e$ . But  $m_\mu$  is known to 10 significant figures, so the prediction uncertainty from input masses is below 0.01 MeV — far smaller than the PDG uncertainty on  $m_\tau$  itself (0.12 MeV). The test of the formula is limited by the  $m_\tau$  measurement, not by the  $m_e$  or  $m_\mu$  inputs.

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## APPENDIX Q: THE PARAMETER REDUCTION — COMPLETE ACCOUNTING

| Step                                 | Parameter Count | What Changed                                                                                                                                                                          | Paper        | Method                                                         | Conditional?                                          |
|--------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------------------------------------|-------------------------------------------------------|
| Standard Model (textbook)            | 19              | Baseline: 3 gauge couplings, 6 Yukawa couplings, Higgs $\mu^2$ and $\lambda$ , 3 CKM angles, 1 CKM phase, $\theta$ QCD, plus 3 neutrino masses and 3 PMNS angles if massive neutrinos | —            | —                                                              | —                                                     |
| After PHYS-7                         | 18              | $\theta$ QCD = 0 derived as ground state                                                                                                                                              | PHYS-7       | Energy minimization of integer-topological system              | No — the ground state argument is unconditional       |
| After PHYS-8                         | 17              | $m_\tau$ derived from $m_e$ , $m_\mu$ via Koide                                                                                                                                       | PHYS-8       | Trigonometric identity at $N = 3$ , $a^2 = 2$                  | <b>Yes</b> — conditional on Koide formula being exact |
| Future: if Koide extends to quarks   | 15 or less      | $m_b$ from $m_d$ , $m_s$ ; $m_t$ from $m_u$ , $m_c$                                                                                                                                   | Hypothetical | Same formula at different amplitude?                           | Highly conditional — quarks don't currently fit       |
| Future: if $M$ or $\theta_0$ derived | 16              | One of $\{m_e, m_\mu\}$ derived from the other + structure                                                                                                                            | Hypothetical | Would require deriving $M$ or $\theta_0$ from deeper principle | Speculative                                           |



| Step                | Parameter Count | What Changed                          | Paper | Method  | Conditional? |
|---------------------|-----------------|---------------------------------------|-------|---------|--------------|
| Theoretical minimum | ?               | All parameters derived from structure | ToE   | Unknown | Unknown      |

**Current status: 19 → 18 (unconditional) → 17 (conditional on Koide exactness at  $0.91\sigma$ ).**

## APPENDIX R: FUTURE $m_\tau$ MEASUREMENTS — WHAT WOULD CONFIRM OR FALSIFY

| Experiment/Facility                | Expected $m_\tau$ Precision | Timeline  | What It Would Show                                                                                                                                       |
|------------------------------------|-----------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Belle II (KEK)                     | $\pm 0.05\text{-}0.08$ MeV  | 2025-2030 | If central value moves toward 1776.97: strengthens Koide to $>2\sigma$ confirmation. If moves below 1776.74: $>2\sigma$ tension with Koide.              |
| FCC-ee (CERN, proposed)            | $\pm 0.01\text{-}0.02$ MeV  | 2040s     | Definitive test. At this precision, Koide prediction of 1776.97 is either confirmed (within 0.02) or falsified ( $>5\sigma$ tension if PDG value holds). |
| Super Tau-Charm Factory (proposed) | $\pm 0.03\text{-}0.05$ MeV  | 2030s     | Intermediate precision. Could confirm or begin to exclude.                                                                                               |
| Current PDG average                | $\pm 0.12$ MeV              | Now       | $0.91\sigma$ — consistent with exact Koide                                                                                                               |

**The falsification threshold:** At the current 0.12 MeV uncertainty, the Koide prediction would be excluded at  $3\sigma$  if  $m_\tau < 1776.61$  MeV or  $m_\tau > 1777.33$  MeV. The current central value 1776.86 is within  $1\sigma$ . Improvement to  $\pm 0.05$  MeV would narrow the  $3\sigma$  window to 1776.82-1777.12, still including the Koide prediction. Improvement to  $\pm 0.02$  MeV would make the test sharp: the  $3\sigma$  window would be 1776.91-1777.03, centered on the Koide prediction.

[[HOWL-PHYS-8-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-8-2026>

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[[HOWL-PHYS-7-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-7-2026>

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